MODULE 08 · DOMAIN 1 - MONITORING, LOGGING, ANALYSIS, REMEDIATION, AND PERFORMANCE OPTIMIZATION

Automated Remediation and Self-Healing

Wiring detection to action - and the newest tools in scope, AWS DevOps Agent and Kiro.

DOMAIN 1 of 5 EXAM WEIGHT 22% MODULE 8 of 35

admin@sthithapragna.com

Automate the fix you already understand. **Investigate** the one you do not. Confusing those two is how automation causes outages.

The remediation chain

01
Detect
An alarm, a Config rule, a CloudTrail event, a Health event.

›

02
Route
EventBridge, or the alarm action itself.

›

03
Decide
Guardrails: is this safe, is it in scope, has it already run?

›

04
Act
SSM Automation, Lambda, Step Functions, an EC2 action.

›

05
Record
An OpsItem, a notification, a CloudTrail entry.

ON THE EXAM

Most designs skip Decide and Record. Skipping Decide is how you get a remediation loop; skipping Record is how nobody learns anything.

Five paths from detection to action

PATH	RIGHT WHEN
Alarm with an EC2 action	Reboot, stop, terminate or recover one instance. No routing needed
Alarm with an Auto Scaling action	The fix is more capacity, or replacing an unhealthy instance
EventBridge to SSM Automation	A known multi-step fix on AWS resources. The workhorse
Config rule to SSM Automation	A resource has drifted from a compliance rule and must be put back
EventBridge to Step Functions	The fix needs waits, branching, or a human approval

ON THE EXAM

Lambda is missing on purpose. It is always available and rarely the best answer when a native path exists.

The simplest self-healing

- **StatusCheckFailed_System with the recover action.** Broken host, same instance, no code anywhere.
- **Auto Scaling health checks** replace an unhealthy instance automatically. EC2 checks by default, or ELB health checks.
- Turn on **ELB health checks in the group** so an instance that is running but not serving traffic is also replaced.
- These fix **symptoms, fast**. Neither one tells you why, so pair them with something that records the event.

> ON THE EXAM
>
> An Auto Scaling group with ELB health checks is self-healing with no automation to write. Reach for it before building anything.

EventBridge to SSM Automation: the workhorse

- The rule matches the event; the **runbook is a native target**. No Lambda function in between.
- Works from a **CloudWatch alarm state change**, a **CloudTrail API event**, or a **Health event**.
- Hundreds of **AWS-owned runbooks** already exist for common fixes, so often there is nothing to write.
- The runbook can carry **rate control and approval steps**, which a Lambda function would have to implement itself.

> ON THE EXAM
>
> "Automatically revert an unauthorised change" is a CloudTrail event to a runbook. This one pattern answers a lot of questions.

Config rules and remediation

- A **Config rule** evaluates resources against a desired configuration and marks them compliant or not.
- A **remediation action** attaches an SSM Automation runbook to the rule, set to **automatic** or **manual**.
- Automatic remediation supports **retries**, with a maximum attempt count over a time window.
- It needs a **remediation execution role** that the runbook assumes. A missing or under-scoped role is the usual failure.

> ON THE EXAM
>
> Config evaluates state, not events. It notices a resource is wrong even if the change that made it wrong was never seen.

Config remediation or EventBridge remediation?

CONFIG WHEN

State has drifted

- Continuous evaluation against a rule
- Catches resources that were already wrong
- Also catches changes made before the rule existed
- Slower: evaluation is periodic or change-triggered

EVENTBRIDGE WHEN

Something just happened

- Reacts to the API call itself
- Near real time, seconds not minutes
- Only sees events after the rule was created
- No compliance history or reporting

THE TIEBREAKER

Within one minute means EventBridge. Compliance reporting, or fixing what is already broken, means Config.

Prevention beats remediation

- **State Manager** re-applies desired state on a schedule, so drift is corrected before anyone notices it.
- **SCPs and IAM** stop the bad change from happening at all. Nothing to remediate.
- **CloudFormation drift detection** reports resources that no longer match their template.
- Remediation is a **detective control with an action attached**. It always runs after the damage.

> ON THE EXAM
>
> If a question offers a preventive option and a remediating one, and both meet the requirement, the preventive one is usually the answer.

Guardrails that keep automation safe

- **Idempotent.** Running the fix twice must be harmless, because it will run twice.
- **Bounded blast radius.** Rate control and error thresholds, so a bad fix stops after five resources rather than five hundred.
- **Approval steps** in a runbook for anything destructive or expensive.
- **Loop protection.** If the fix can re-trigger the detection, you have built a machine that bills you all weekend.

> ON THE EXAM
>
> The scariest failure is not the remediation that does not run. It is the one that runs correctly, repeatedly, against the wrong thing.

When not to automate

- **The root cause is unknown.** Restarting something that is failing for an unknown reason destroys the evidence.
- **The fix is destructive and irreversible.** Deleting data, terminating a stateful node, changing DNS.
- **The condition fires during deployments.** Automation that fights a deployment will win, and that is worse.
- In these cases automate the **investigation and the notification**, and leave the action to a human.

> ON THE EXAM
>
> Automating the gathering of evidence is nearly always safe. Automating the change is the part that needs judgement.

AWS Health Dashboard events

- Added to the in-scope service list in **exam guide v1.1, June 2026**.
- Publishes **scheduled changes, degradations and account notifications** as events on the default EventBridge bus.
- Examples: an **EC2 instance scheduled for retirement**, a certificate about to expire, an underlying host degradation.
- Organizational view aggregates health events for **every account** in the organisation.

> ON THE EXAM
>
> "Act before AWS retires the instance" is a Health event to a runbook. No alarm or metric will tell you that in advance.

AWS DevOps Agent: what it actually is

- An **autonomous operations agent**, generally available **31 March 2026**, built on **Amazon Bedrock AgentCore**.
- Three modes: **incident investigations**, **proactive reliability evaluations**, and **on-demand SRE tasks**.
- Triggered by a **webhook**: a CloudWatch alarm, or a third-party alert from PagerDuty, Datadog or ServiceNow.
- Investigates **AWS, Azure and on-premises** workloads. Non-AWS reach is through the Model Context Protocol.

> ON THE EXAM
>
> This is why Amazon Bedrock joined the in-scope list in v1.1. DevOps Agent is built on it, and the two arrived together.

How a DevOps Agent investigation runs

- It builds a **topology of your application** and correlates metrics, logs, traces, deployments and code changes.
- It forms **hypotheses**, tests each against telemetry, and narrows to a probable root cause with evidence.
- It produces a **mitigation plan for human approval**. It does not execute changes on its own.
- Every step is written to an **immutable audit journal** that the agent cannot edit, and it integrates with CloudTrail.

> ON THE EXAM
>
> The third point is the exam-relevant one. DevOps Agent shortens diagnosis, not approval. A human still decides.

Adopting DevOps Agent without losing control

- An **Agent Space** is the unit of setup. Admins configure it in the console; operators work in the DevOps Agent web app.
- Start in **recommendation-only mode**. The agent proposes and humans approve everything, so you can measure it before trusting it.
- Connects to **CloudWatch** plus third-party tools: Datadog, Dynatrace, New Relic, Splunk, GitHub, GitLab, ServiceNow, Slack.
- Billed **per second of agent working time**, not per seat. New customers get a free trial.

> ON THE EXAM
>
> The webhook is the integration point. A CloudWatch alarm action pointed at it starts an investigation automatically.

Runbook or DevOps Agent?

SSM AUTOMATION RUNBOOK

You know the answer

- Deterministic: this condition, that fix
- Runs in seconds, unattended
- Auditable and testable in advance
- Useless against a cause you did not anticipate

DEVOPS AGENT

You do not know the answer

- Correlates across tools to find a root cause
- Handles novel and multi-service failures
- Produces a plan, then waits for approval
- Diagnosis in minutes, not the fix itself

THE TIEBREAKER

Known failure, known fix, act now: runbook. Unknown cause, many services, need the reason: DevOps Agent.

Kiro, and being honest about where it fits

- An **agentic IDE**, generally available November 2025. Spec-driven: requirements, design and tasks as real files.
- **Agent hooks** run actions on events such as a file save or a commit. **Steering files** encode team standards persistently.
- For operations, it is where you **author** runbooks, IaC and Lambda functions - not where they run.
- The **Kiro Autonomous Agent** is still preview. Do not assume it in an answer.

> ON THE EXAM
>
> The exam guide lists Kiro under automating remediation. It builds the automation; SSM, EventBridge and Config run it.

Four AI services, and why they all arrived at once

SERVICE	WHAT IT IS FOR, IN ONE LINE
Amazon Bedrock	The foundation. DevOps Agent is built on Bedrock AgentCore
AWS DevOps Agent	Investigates incidents and proposes mitigations. Operations
Kiro	Agentic IDE where you author automation. Development
AWS Security Agent	Penetration testing and code scanning. Security. Being renamed to Continuum
Amazon Q Developer	Being retired. New sign-ups blocked 15 May 2026, IDE plugins end 30 April 2027

ON THE EXAM

Q Developer inside the AWS console and first-party experiences is NOT affected. Only the IDE plugins and paid subscriptions are retiring.

A compliance rule requires that every new EBS volume is encrypted. Several unencrypted volumes already exist from before the rule was written, and developers occasionally create more. The security team must be able to show auditors a compliance history for every volume.

Remediation should happen without a human, but the team accepts it may take a few minutes rather than seconds.

What should the team implement?

A An EventBridge rule on CreateVolume, targeting a Lambda function that checks and fixes encryption

B An AWS Config rule with an automatic SSM Automation remediation action

C A State Manager association running a Run Command document hourly

D An SCP denying ec2:CreateVolume when encryption is not specified

◆ PAUSE AND DECIDE

What should the team implement?

A EventBridge on CreateVolume plus Lambda

Only sees volumes created after the rule exists, so the pre-existing ones stay wrong. It also produces no compliance history.

B Config rule with automatic remediation

Evaluates state, so it finds the old volumes too, remediates without a human, and produces exactly the compliance history auditors want.

C State Manager association

State Manager acts on managed nodes through documents run on instances. It is not a mechanism for evaluating EBS volume properties.

D SCP denying unencrypted volumes

Genuinely good prevention, and it does nothing about the volumes that already exist. It also produces no compliance report.

THE SIGNAL

Two clauses decide it: existing resources must be fixed, and auditors need history. Only state-based evaluation gives you both.

> An application on EC2 becomes unresponsive roughly once a fortnight. Restarting the process always clears it. Nobody knows why it happens, and the team has been restarting it manually at whatever time it occurs.
>
> The engineering manager proposes an EventBridge rule that automatically restarts the process whenever the alarm fires, so nobody is woken up again.

What is the best response to this proposal?

A Implement it as proposed. The fix is known and reliable

B Automate evidence collection and notification first, and restart only after the root cause is understood

C Replace the instance with an Auto Scaling group of one and let health checks handle it

D Increase the alarm's datapoints to alarm so it fires less often

◆ PAUSE AND DECIDE

What is the best response to this proposal?

A. Implement as proposed

The fix is known; the cause is not. An automatic restart destroys the state that would have explained a fortnightly failure.

B. Collect evidence first, restart later

Capture a thread dump, memory and logs automatically, notify, then restart once the cause is understood. Nobody is woken and nothing is lost.

C. Auto Scaling group of one

Automated restarting with extra steps. It hides the fault behind a replacement and still explains nothing.

D. Raise datapoints to alarm

Makes detection slower and less sensitive. The application still hangs; you just hear about it later.

THE SIGNAL

"Nobody knows why it happens" is the clause. A reliable fix for an undiagnosed fault is a way of never diagnosing it.

> During a payment outage, four engineers spent three hours correlating CloudWatch metrics, RDS Performance Insights, application logs and a GitHub deployment history before finding that a schema migration had removed an index. Similar multi-service investigations happen most months.
>
> The company wants to shorten diagnosis. Compliance requires that a human approves every production change, and that the investigation is auditable.

What should the company adopt?

A. An SSM Automation runbook triggered by the payment latency alarm, which restarts the service

B. AWS DevOps Agent, triggered by the alarm, producing a root cause and a mitigation plan for approval

C. CloudWatch anomaly detection on every metric in the payment stack

D. Kiro, to generate remediation code for each failure as it occurs

◆ PAUSE AND DECIDE

What should the company adopt?

A Runbook that restarts the service

A deterministic fix for a cause nobody identified in advance. A restart would not have replaced a dropped index.

B DevOps Agent investigation

Correlates telemetry, logs and deployment history exactly as the engineers did, proposes a plan for approval, and journals every step immutably.

C Anomaly detection everywhere

Improves detection, which was never the problem. They knew within minutes that it was broken; they needed to know why.

D Kiro to generate remediation code

Kiro is a development environment. It writes automation ahead of time; it is not an incident response tool.

THE SIGNAL

Every clause fits: multi-tool correlation, novel causes, approval required, auditable. That is the DevOps Agent shape exactly.

AWS notifies an account that 30 EC2 instances are scheduled for retirement in 14 days because their underlying hardware is being decommissioned. The team only discovered this by chance, reading the console.

They want future notifications of this kind to automatically raise a tracked work item and start a runbook that stops and starts each affected instance during the next maintenance window.

What should the team configure?

A. A CloudWatch alarm on StatusCheckFailed_System with the recover action

B. An EventBridge rule on AWS Health events, targeting an SSM Automation runbook and creating an OpsItem

C. An AWS Config rule detecting instances scheduled for retirement

D. A daily Lambda function that calls DescribeInstanceStatus and emails the team

◆ PAUSE AND DECIDE

What should the team configure?

A. Status check alarm with recover

Fires when the host has already failed. Retirement is a scheduled future event, so nothing is failing yet and no alarm exists.

B. EventBridge on AWS Health events

Health publishes scheduled changes to the default bus. One rule gives both the tracked OpsItem and the runbook.

C. Config rule for retirement

Config evaluates resource configuration. A retirement notice is an AWS-side event, not a property of the instance.

D. Daily Lambda plus email

Polling rebuilt by hand, and an email is not a tracked work item or an automated fix. It answers neither requirement.

THE SIGNAL

Nothing is broken yet. That single observation eliminates every metric and alarm answer, because there is no metric for a future event.

The five things to carry forward

- **EventBridge to an SSM Automation runbook** is the default remediation pattern. No Lambda in the middle.
- **Config for drifted state, EventBridge for fresh events.** Existing resources or compliance history means Config.
- **Prefer prevention**, unless the scenario says resources are already wrong.
- **Unknown cause means investigate, not restart.** Automate evidence gathering; it is always safe.
- **DevOps Agent proposes, a human approves.** Runbooks act. AWS Health warns before anything breaks.

> ON THE EXAM
>
> Next: compute and EBS performance - instance families, burst credits, volume types and placement groups.